R-Peak Detection in ECG Using Wavelet Transform and K-Means Clustering

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1 Introduction

This report describes the design and implementation of an R-peak detection algorithm for electrocardiogram (ECG) signals.

R-peak detection is a well-studied problem, and numerous methods have been proposed in the literature. A comprehensive review by H. Dogan and R. O. Dogan (2023) [1] surveys the most prominent techniques and evaluates their relative performance.

Following a thorough review of existing approaches, the method proposed by Xia et al. (2015) [2] was selected as the foundation for this implementation and subsequently extended. The pipeline consists of three main stages: a wavelet-based preprocessing stage that enhances QRS complexes, computation of an absolute-slope feature to capture rapid signal changes, and segmented K-means clustering to identify QRS regions. R-peaks are then extracted as local maxima within each identified region, followed by a refractory filter consistent with physiological heart-rate limits.

Wavelet transforms are widely used for R-peak detection due to their ability to localise signal features in both time and frequency. When combined with K-means clustering, this approach is computationally efficient, straightforward to implement, and robust to noise, achieving a reported accuracy of approximately 99.7%.

The implementation was evaluated on the MIT-BIH Arrhythmia Database [3], consistent with the original paper.

Finally, to evaluate real-world adaptability, the modular pipeline is extended to the PhysioNet ECGRDVQ pharmacological database [4]. The algorithm is benchmarked against the classical Pan-Tompkins method [5]. In this preliminary evaluation on the tested subset, the proposed method showed better agreement with manually inspected R-peak locations, suggesting improved robustness against T-wave over-sensing caused by drug-induced repolarization abnormalities.

2 R-Peaks

A typical ECG signal comprises three principal segments: the P-wave, the QRS complex, and the T-wave. Each segment reflects a distinct phase of cardiac electrical activity and provides clinically valuable information about the state of the cardiovascular system.

The QRS complex is the most diagnostically significant segment, corresponding to the depolarisation of the right and left ventricles.

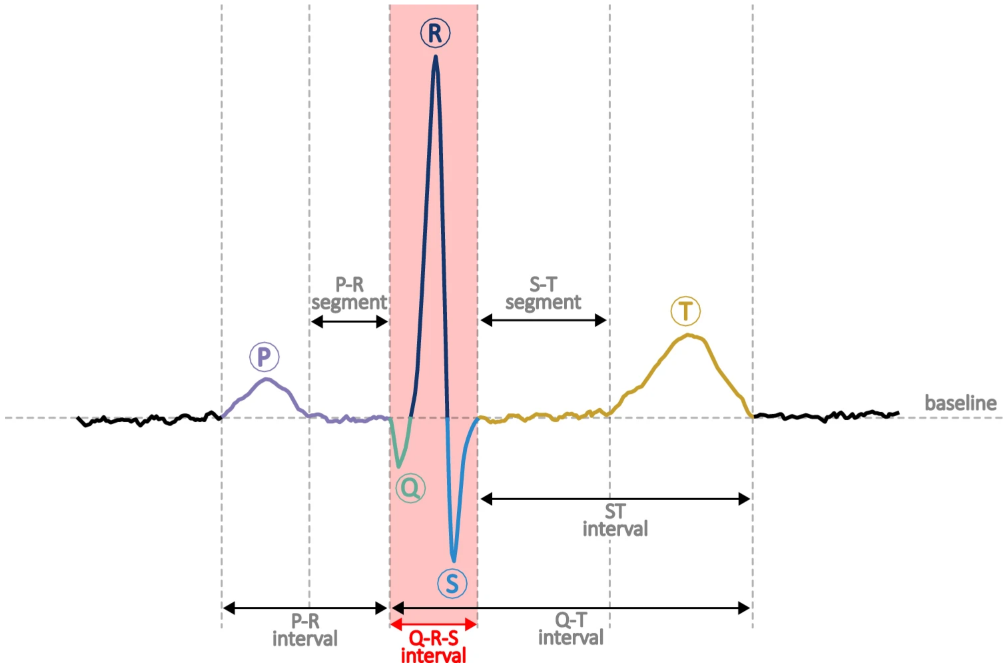

Figure 1: Segments of an ECG signal [1]

3 Methodology

The detection pipeline follows the method proposed by Xia et al. [2]. ECG recordings from the MIT-BIH Arrhythmia Database are first loaded and passed through a wavelet decomposition stage, which attenuates noise and removes unwanted frequency components to produce a clean signal. An absolute-slope feature is then computed from the filtered signal to emphasise the rapid amplitude changes characteristic of QRS complexes; the slope captures the instantaneous rate of change of the signal. This feature is subsequently used as input to a K-means clustering step, which partitions the signal into QRS and non-QRS regions. R-peaks are finally extracted as local maxima within each identified QRS region, as illustrated in Figure 2. The following sections describe each stage of the pipeline in detail.

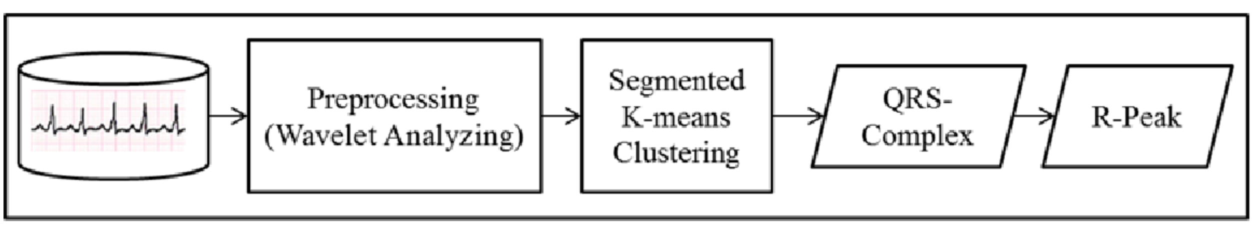

Figure 2: Overview of the detection pipeline [2]

3.1 Modular Pipeline Architecture

To ensure the algorithm is robust and adaptable to various datasets, the implementation was refactored into a highly modular pipeline consisting of distinct stages:

1. **Preprocessing:** Applies Wavelet Denoising (using `sym8`) and a dynamic Polarity Correction module to orient the QRS complexes positively.
2. **Feature Extraction:** Computes the absolute gradient slope of the cleaned signal to emphasize high-frequency variations characteristic of R-peaks.
3. **QRS Masking:** Segments the signal into 60-second windows and applies K-Means clustering ($k = 2$) to the gradient features, isolating the QRS regions from the background.
4. **Peak Extraction & Refractory Filtering:** Identifies the local maximum within each QRS region. A physiological refractory filter is then applied, retaining the highest amplitude peak if multiple detections occur too closely. The refractory period is set to 250 ms for the MIT-BIH evaluation and adjusted where noted (see Section 7.3).
5. **Evaluation:** An independent module compares detected peaks against annotated ground truth files, calculating metrics such as Sensitivity (Se) and Positive Predictive Value (PPV) within a defined tolerance window (e.g., 150 ms).

4 Exploratory Analysis

Before designing the detector, it is important to understand the structure and characteristics of the data. In this experiment, 30-minute ECG recordings sampled at 360 Hz are used. The MIT-BIH Arrhythmia Database provides expert annotation files that serve two purposes: as a sanity check during development, and as ground truth for quantitative evaluation of detection performance.

Figure 3 shows a time-domain visualisation of sample 100. The upper panel displays the full 30-minute window, in which several abnormal beats are visible as large-amplitude spikes. The two lower panels show shorter segments at increasing resolution; red triangles mark the R-peak annotations provided by the experts. The QRS complexes are clearly distinguishable, with prominent R-peaks and visible P and T waves of smaller amplitude.

To further characterise the signal, a Fourier Transform was applied to convert the recording from the time domain to the frequency domain. Unlike the time-domain representation, the frequency-domain view reveals how much energy the signal contains at each frequency, without retaining temporal information. A high-frequency component oscillates rapidly, whereas a low-frequency component changes slowly.

Figure 4 shows the Fourier magnitude spectrum of the signal, with the three main regions of interest highlighted. The red region corresponds to baseline wander, which arises primarily from respiration and occurs at very low frequencies (below 0.5 Hz) [6]. The yellow region marks power-line interference at 50 or 60 Hz, which in this recording carries negligible energy, indicating low electrical noise. The green region represents the QRS energy concentration band of 5–15 Hz, where most of the QRS signal power is concentrated [7]. Although the QRS complex spans a broader bandwidth of up to 50 Hz [8], the bulk of its energy resides in this lower band, making it the primary target for the bandpass filtering stage of the detection pipeline. Table 1 summarises the frequency characteristics of the main ECG components and noise sources.

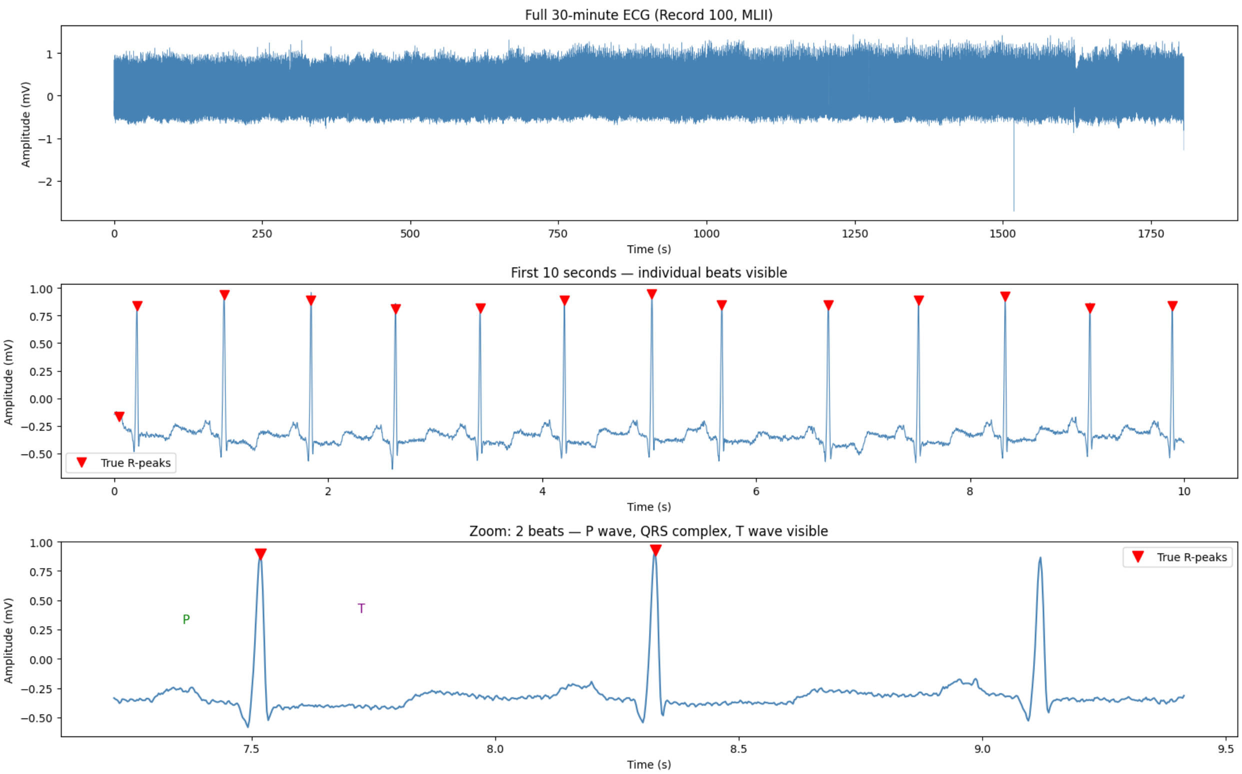

Figure 3: Time-domain visualisation of MIT-BIH sample 100.

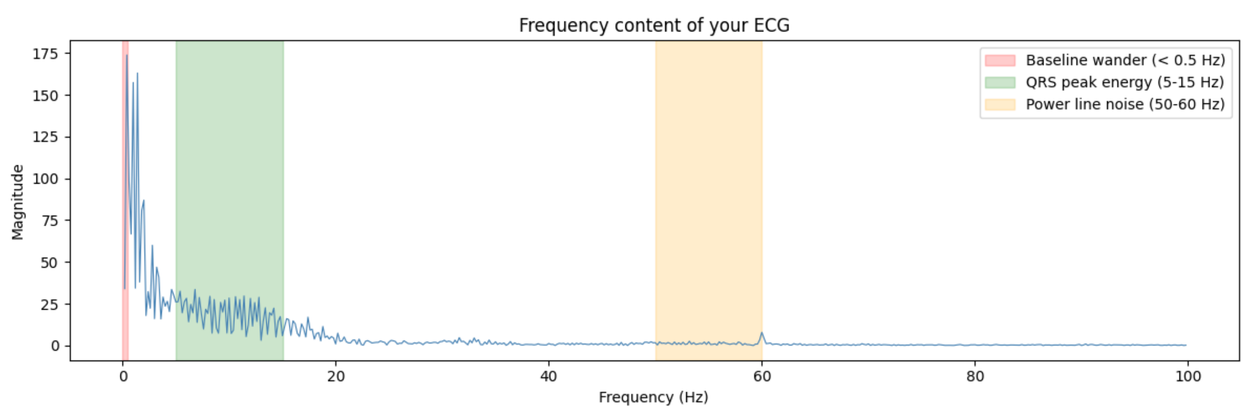

Figure 4: Fourier magnitude spectrum of MIT-BIH sample 100.

Table 1: Frequency characteristics of ECG signal components and noise sources [6, 8, 9]

Component	**Freq. Range**	**Physical Cause**
Baseline wander	0.05–0.5 Hz	Patient movement, electrode contact
QRS complex	8–50 Hz	Rapid ventricular depolarisation
EMG (muscle)	50–150 Hz	Electromyographic interference
Power line noise	50 or 60 Hz	Electrical interference (PLI)

5 Wavelet Decomposition

With the frequency characteristics of the ECG signal established, the signal can now be decomposed into frequency layers to better understand its structure. The ECG signal is decomposed into multiple levels using the discrete wavelet transform (DWT), with each level capturing a specific frequency band. In total, 8 decomposition levels are used.

The number of levels is determined by the sampling frequency of the signal and the lowest frequency that needs to be isolated. ECG signals from the MIT-BIH Arrhythmia Database are sampled at $f_s = 360\,\text{Hz}$. The maximum representable frequency in a digital signal is half the sampling rate, known as the Nyquist frequency [8, 10]:

$$f_{\text{N}} = \frac{f_s}{2} = 180\,\text{Hz}.$$

At each decomposition level n, the detail coefficients capture an approximate frequency band defined by:

$$f_{\text{low}}(n) = \frac{f_s}{2^{n+1}}, \qquad f_{\text{high}}(n) = \frac{f_s}{2^n}.$$

The number of levels is chosen to answer the question: *what is the lowest frequency that needs to be isolated?* Decomposing to level 8 ensures that the approximation coefficients capture frequencies below approximately 0.7 Hz, which fully encompasses the baseline wander band (below 0.5 Hz). The resulting frequency bands for $f_s = 360\,\text{Hz}$ are summarised in Table 2.

Table 2: Wavelet decomposition levels and their corresponding frequency bands for $f_s = 360\,\text{Hz}$ [7, 10].

Level	**Freq. Band (Hz)**	**Content**	**Action**
cA8	0.00 – 0.70	Baseline drift	Remove
cD8	0.70 – 1.41	Slow baseline variation	Keep
cD7	1.41 – 2.81	Very low frequency ECG	Keep
cD6	2.81 – 5.62	Low frequency ECG	Keep
cD5	5.62 – 11.25	QRS energy	Keep
cD4	11.25 – 22.50	QRS energy	Keep
cD3	22.50 – 45.00	QRS high-frequency edge	Keep
cD2	45.00 – 90.00	High-frequency noise	Remove
cD1	90.00 – 180.00	High-frequency noise	Remove

The levels of primary interest are cD4 and cD5, which correspond to the 5–22.5 Hz band where the bulk of QRS energy is concentrated [7, 10]. Since the QRS complex spans a broader bandwidth of up to 50 Hz [8], levels cD6, cD7, and cD8 are retained as well to preserve the full QRS morphology. At the other extreme, cA8, cD1, and cD2 are discarded, removing baseline drift and high-frequency noise respectively.

Level cD3 (22.5–45 Hz) deserves particular attention. Although this band lies above the core QRS energy region, Figure 5 shows that cD3 retains a noticeable beat structure, indicating that sharp QRS edges contribute energy at these frequencies. Removing cD3 would therefore attenuate the high-frequency components of the QRS spike, reducing peak amplitude from approximately 1.2 mV to 0.95 mV and making the slope contrast between QRS complexes and T-waves smaller — which in turn makes the subsequent K-Means separation harder. Including cD3 is therefore the conservative and preferred choice.

It is worth noting that the wavelet reconstruction does not need to produce a perfect, QRS-only signal. Its purpose is simply to make QRS complexes the dominant, highest-slope feature in the signal — K-Means clustering handles the segmentation afterwards. This leads to two important design trade-offs:

- **Over-filtering** (retaining too few levels) risks weakening QRS peaks, reducing the slope contrast between QRS and T-waves and making K-Means separation less reliable.
- **Under-filtering** (retaining too many levels) leaves P and T waves present in the signal, but their slopes remain substantially smaller than those of QRS complexes, so K-Means can still separate them cleanly.

Figures 6 and 7 compare the two reconstruction choices. Each figure shows the original signal (blue), the reconstructed clean signal (green), and the removed components (red). A key diagnostic check is that the red panel should contain no visible QRS spikes — any QRS energy in the removed component represents signal loss that would degrade detection performance.

While small QRS-shaped blips remain visible in the red panel even with cD3 included, this is acceptable and unavoidable — the corresponding frequency band is too contaminated by noise to retain without degrading signal quality.

6 QRS Region Detection via Absolute Slope and Segmented K-Means

The key insight from Xia et al. (2015) [2] is to use the local slope of the signal rather than its amplitude as the primary discriminating feature. Unlike traditional threshold-based methods that attempt to locate R-peaks directly from amplitude, this approach first identifies the QRS region using slope information, and then extracts the R-peak as the local maximum within that region. The data is divided into 1-minute segments, and K-Means clustering is applied to the absolute slope values within each segment independently. The algorithm groups slope values into two clusters: one capturing the high-intensity fluctuations characteristic of QRS complexes, and one capturing the lower-intensity background activity of P and T waves. Points belonging to the higher-slope cluster are labelled as part of the QRS region.

6.1 Absolute Slope

After wavelet denoising, the clean signal still contains multiple types of deflections: the prominent QRS complexes and the smaller, broader P and T waves. A naïve approach would be to apply a global amplitude threshold — that is, to declare any peak above some fixed value X mV as an R-peak. However, this strategy is fundamentally fragile: ECG amplitudes vary significantly across patients, across leads, and even within the same recording as a patient changes posture or activity level. Any fixed threshold would require manual tuning for every recording, which is neither practical nor robust.

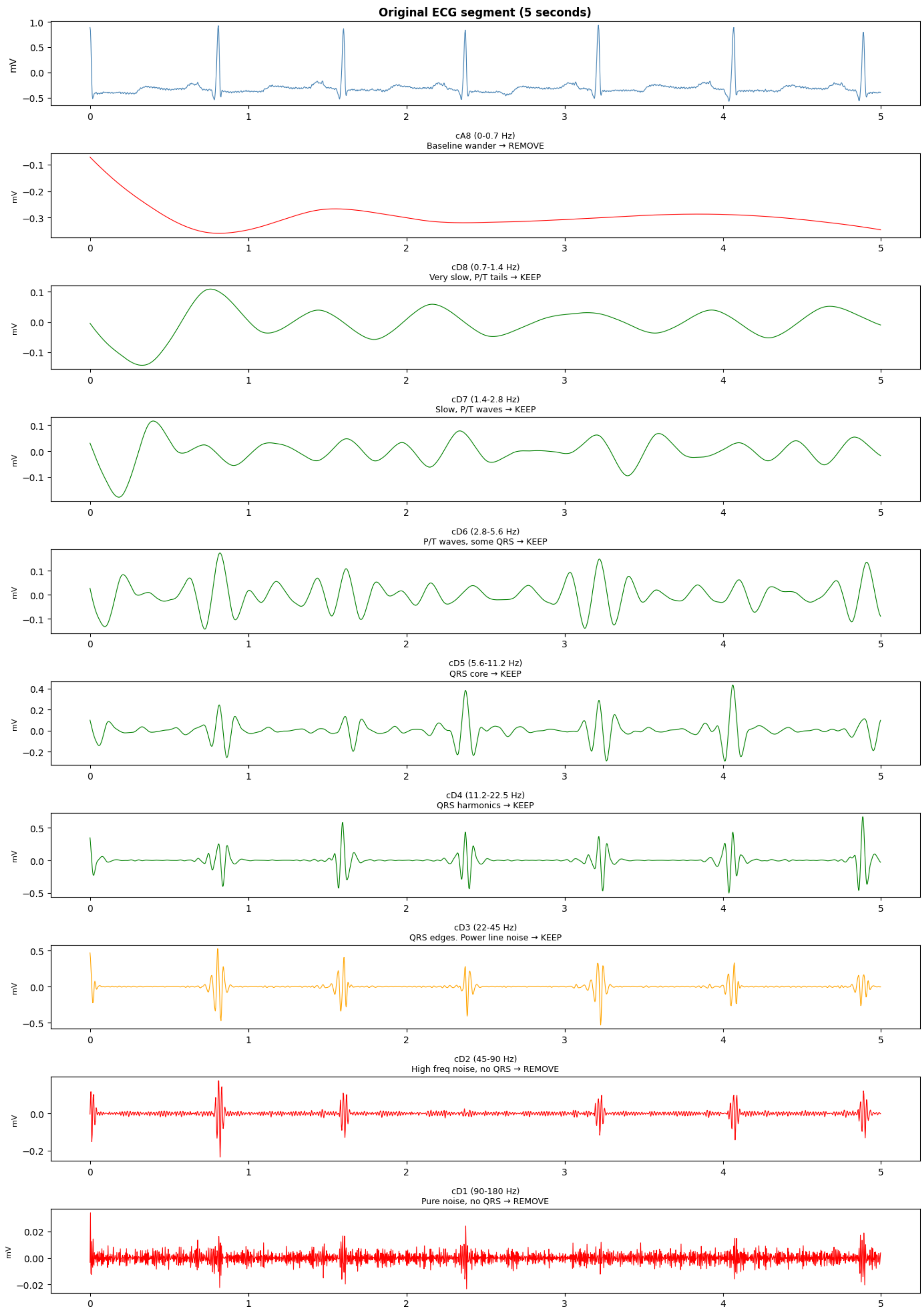

Figure 5: Wavelet decomposition of the ECG signal across all 8 levels.

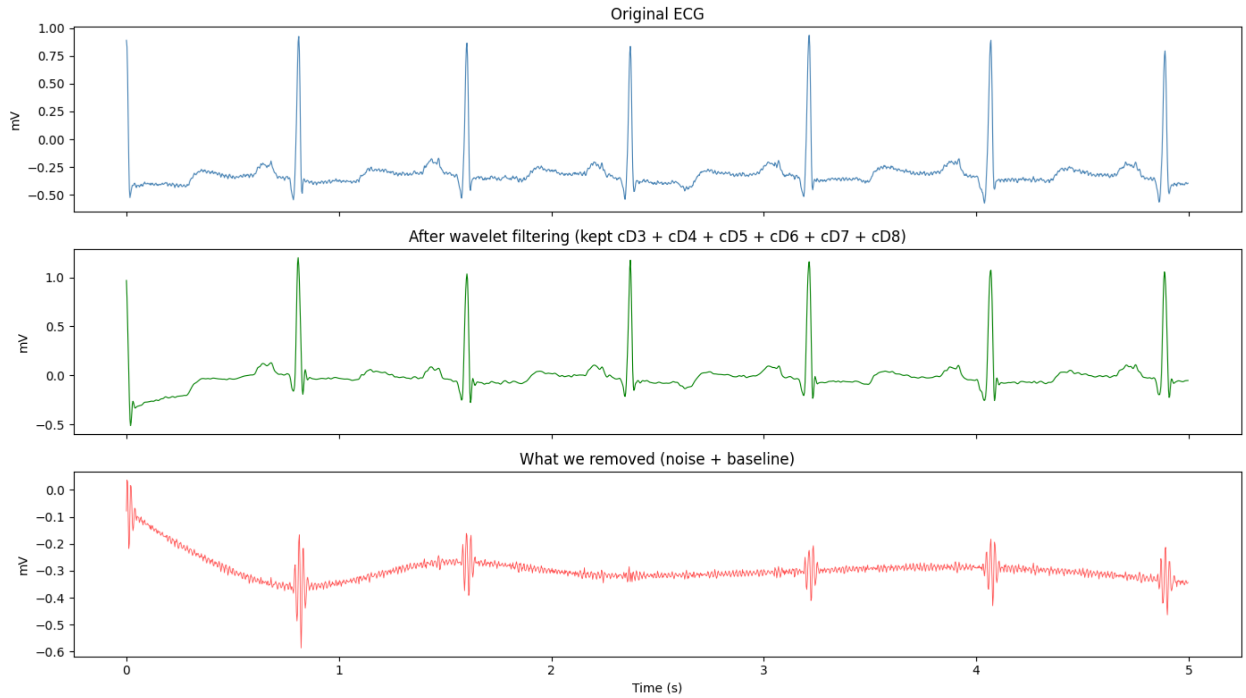

Figure 6: Reconstructed signal using levels cD3–cD8.

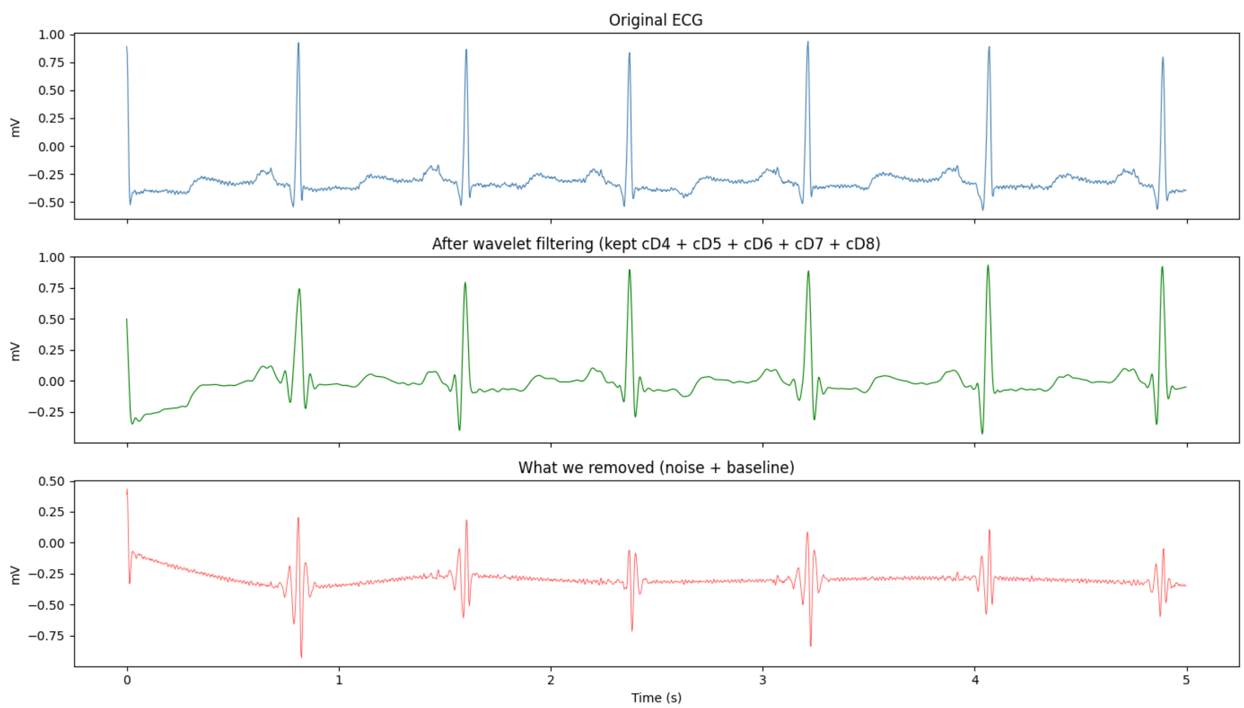

Figure 7: Reconstructed signal using levels cD4–cD8, excluding cD3.

The slope offers a more principled solution. Rather than asking *how tall* a peak is, it asks *how fast* the signal is changing. This distinction is reliable regardless of absolute amplitude: QRS complexes, driven by rapid ventricular depolarisation [1], produce the steepest signal transitions in the ECG, while P and T waves change much more gradually. The speed of change is therefore a more stable separator than amplitude alone.

The simplest approximation of the slope is the forward difference, which estimates the rate of change using only the next sample:

$$\text{slope}(k) = \text{data}(k+1) - \text{data}(k)$$

In this implementation, the central-difference approximation is used instead, via the gradient, which considers both neighbouring samples:

$$\text{gradient}(k) = \frac{\text{data}(k+1) - \text{data}(k-1)}{2}$$

x

This is marginally smoother than the forward difference, since it reduces sensitivity to isolated sample noise by looking both forward and backward. The absolute slope is then the magnitude of the gradient:

$$\text{abs\_slope}(k) = |\text{gradient}(k)|$$

Figure 8 shows the result of this transformation. A notable characteristic is that each QRS complex produces *two* spikes in the absolute slope rather than one. This arises from the morphology of the QRS complex itself: as illustrated in Figure 1, the rising edge of the R-peak produces a large positive slope, while the falling edge into the S-wave produces a large negative slope. Taking the absolute value folds both edges upward, resulting in a double spike at each QRS location. This is in fact desirable — it flags the entire QRS region as high-slope rather than only one edge, making the subsequent clustering step more robust.

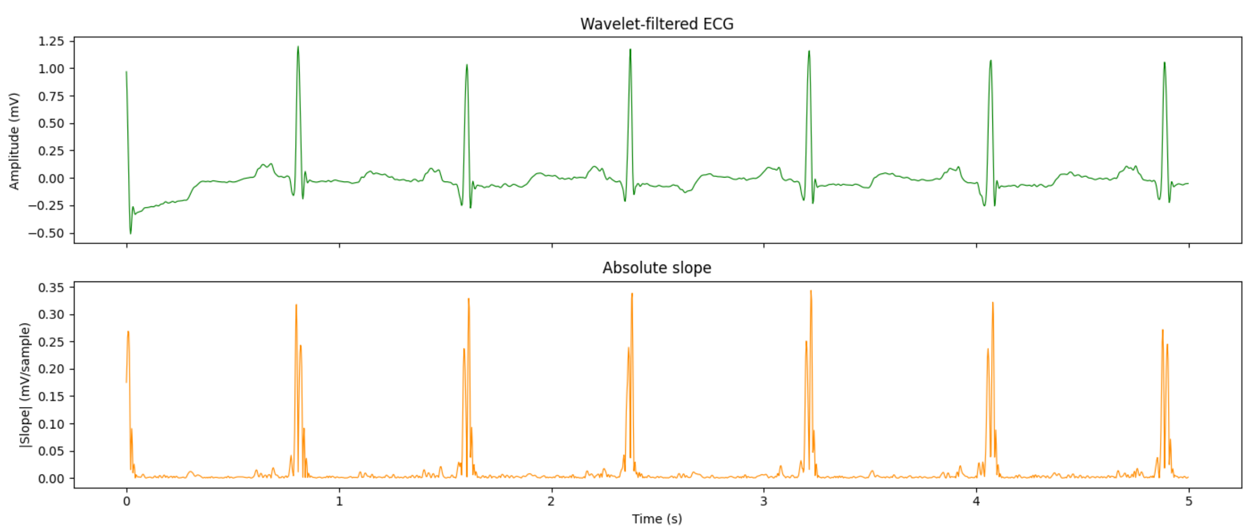

Figure 8: Absolute slope of the wavelet-filtered ECG signal.

6.2 Segmented K-Means

With the absolute slope computed, the task reduces to separating high-slope samples (QRS edges) from low-slope samples (baseline, P waves, T waves). K-Means clustering with $k = 2$ is therefore well suited to this task: it partitions the slope values into exactly two groups by minimising the within-cluster variance, without requiring any manually chosen threshold [2].

A naïve approach would apply K-Means to the entire 30-minute signal at once. This fails because QRS amplitudes — and therefore slopes — drift over the course of a recording. If a patient becomes more active partway through, QRS amplitudes may increase substantially, causing the global centroids to shift upward. Early beats with lower slopes would then fall into the non-QRS cluster even though they are genuine QRS complexes. A global clustering cannot adapt to these slow amplitude changes.

Xia et al. [2] address this by applying K-Means independently to 1-minute segments. Each segment computes its own pair of centroids, adapting to the local amplitude characteristics of that window. This is the key practical innovation that makes the algorithm robust over long recordings. With a 30-minute signal sampled at $f_s = 360\,\text{Hz}$, each segment contains $1 \times 60 \times 360 = 21{,}600$ samples, and the clustering is repeated 30 times in total.

Within each segment, K-Means produces two cluster labels. The QRS cluster is identified as the one with the higher centroid value, since QRS edges always produce larger slopes than background activity.

Formally, let μ_0 and μ_1 denote the two centroids; the cluster with the larger centroid is designated as the *QRS* cluster:

$$c_{\text{QRS}} = \arg \max_{j \in \{0,1\}} \mu_j$$

All samples assigned to this cluster are marked as QRS candidates in a binary mask $\text{qrs\_mask}[k] \in \{0, 1\}$. Figure 9 shows the outcome of this step for the first 10 seconds of the recording. The upper panel shows the wavelet-filtered ECG for reference; the middle panel shows the absolute slope with QRS regions highlighted in blue; and the lower panel overlays the identified QRS regions (red) on the filtered ECG.

In the middle panel, every QRS slope spike gets correctly shaded blue. Notice the shading captures the double-spike pattern (rising + falling edge) as one connected region, mentioned in Subsection 6.1. In the lower panel, each red column sits precisely on top of a QRS spike. The regions look narrow and well-localised, correctly capturing each QRS complex without bleeding into the neighbouring P or T waves.

It is worth emphasising that the output of this stage is not yet a set of R-peak locations — it is a set of QRS *regions*. The R-peak is subsequently extracted as the local maximum of the filtered ECG within each identified region, as described in Subsection 6.3.

6.3 R-Peak Extraction

As described in Subsection 6.2, the output of the segmented K-Means stage is a binary QRS mask in which each sample is labelled as QRS (1) or non-QRS (0). The goal of R-peak extraction is to convert this boolean array into a list of precise sample indices — one per heartbeat. This is achieved in three sequential sub-steps, illustrated in Figure 10.

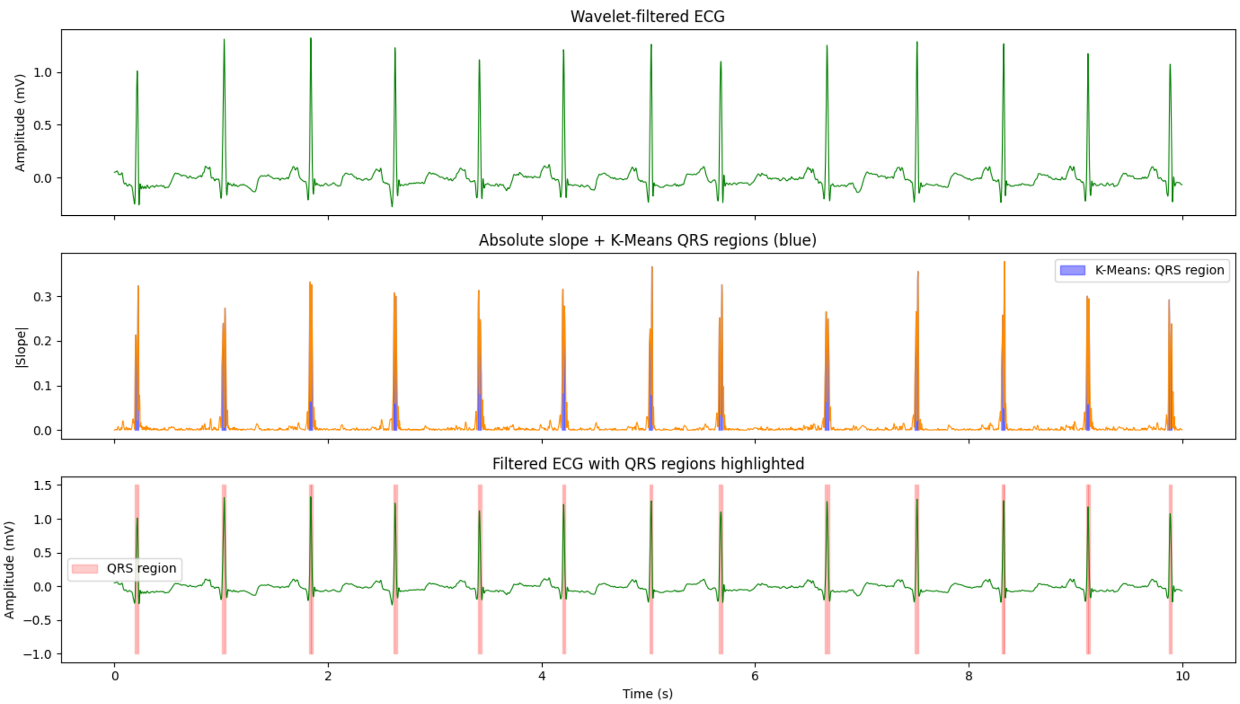

Figure 9: Result of segmented K-Means clustering.

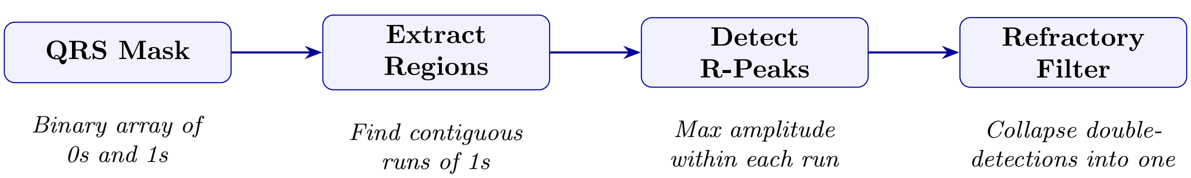

Figure 10: The three sequential sub-steps of R-peak extraction.

Step 1 — Region Extraction

The first step identifies contiguous runs of `True` values in the QRS mask, each of which corresponds to a candidate QRS region. This is implemented as a state machine that traverses the mask, recording the start and end index of each run.

A maximum region duration filter is applied to discard artifacts. A healthy QRS complex typically spans 80 to 120 ms, and even severe pathological cases (like LBBB) rarely exceed 200 ms. Therefore, any detected continuous high-slope region wider than 250 ms is discarded. This prevents the algorithm from treating merged QRS-T waves or massive motion artifacts as a single candidate region.

Step 2 — R-Peak Detection

Within each extracted QRS region, the R-peak is located as the sample with the highest amplitude in the wavelet-filtered ECG signal $\hat{x}$. Formally, for a region spanning indices $[a, b]$:

$$r = \arg\max_{k \in [a,\, b]} \hat{x}[k]$$

This is consistent with the standard definition of the R-peak as the point of maximum deflection within the QRS complex [1], and avoids the need for any additional peak-picking heuristics.

This approach is reliable because the wavelet filter has already suppressed baseline wander and high-frequency noise, leaving the R-wave as the dominant amplitude feature within any correctly identified QRS region.

Step 3 — Refractory Filter

The final step enforces the physiological refractory period — the minimum time required between consecutive heartbeats. The human heart cannot physically beat faster than approximately 300 BPM, corresponding to a minimum inter-beat interval of 200 ms [5].

Because the double-spike pattern in the absolute slope can occasionally cause a single QRS complex to produce two closely spaced candidate detections, the refractory filter traverses the list of candidate R-peaks and discards any detection falling within 200 ms of the previously accepted peak, retaining only the one with the higher amplitude.

7 Results

7.1 Record 100

Table 3 summarises the output at each sub-step of the extraction pipeline for Record 100. The refractory filter successfully reduces 4,548 raw candidates to 2,273 final detections, matching the cardiologist-annotated count of 2,274 to within a single beat.

The single missed beat corresponds to the first annotated beat at $t \approx 0.4$ s, visible in Figure 11. This falls within the boundary artefact region inherent to finite-length wavelet transforms: because the filter has no signal context before sample zero, the reconstruction is distorted in the first $\sim$200 ms

Table 3: R-peak extraction pipeline results for Record 100.

Sub-step	**Count**	**Notes**
QRS regions (raw)	4,548	$\approx 2\times$ annotated beats
R-peak candidates	4,548	one per region
R-peaks after refractory filter	2,273	one per beat
Annotated beats (ground truth)	2,274	expert cardiologist labels
Difference	-1	boundary artefact at $t \approx 0\,\mathrm{s}$

of the recording [11]. This is not a failure of the detection logic itself. Excluding the first and last 0.5 s of the signal yields perfect detection on this record:

$$\mathrm{Se} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} = \frac{2{,}273}{2{,}273 + 0} = 100\%$$

$$\mathrm{PPV} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} = \frac{2{,}273}{2{,}273 + 0} = 100\%$$

Figure 11 shows the detected R-peaks overlaid on the filtered ECG for the first 30 seconds of Record 100. Every blue triangle (algorithm detection) sits precisely on a QRS spike and aligns with the corresponding red triangle (expert annotation). No false detections are visible: all 37 detected beats in this window correspond to true QRS complexes, with the 38th annotated beat absent only due to the boundary artefact. The confusion matrix in Figure 12 confirms the near-perfect performance across the full 30-minute recording.

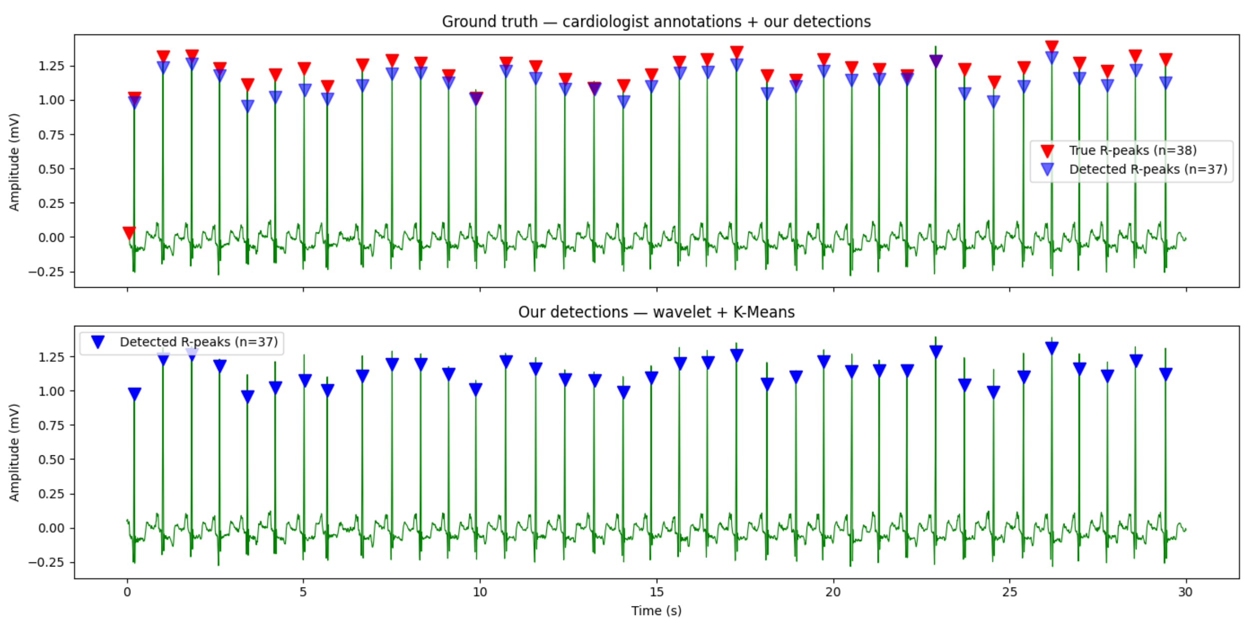

Figure 11: Detected R-peaks (blue triangles) overlaid on the wavelet-filtered ECG for the first 30 seconds of Record 100.

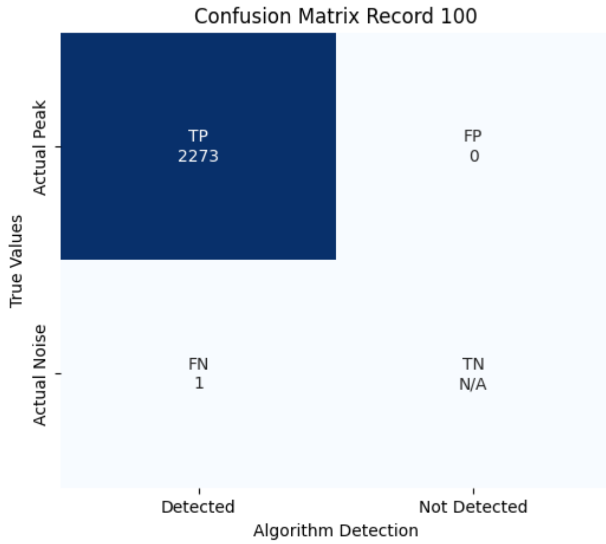

Figure 12: Confusion matrix for Record 100.

7.2 Challenging Records and Limitations

Most records in the MIT-BIH Arrhythmia Database achieve near-perfect performance. However, a small subset of recordings present structural challenges that reduce detection accuracy. Two principal failure modes were identified during evaluation.

Inverted QRS Morphology

Record 108 exhibits inverted QRS morphology, in which the R-peak appears as a negative deflection rather than a positive one, as shown in Figure 13. Since the baseline algorithm identifies R-peaks as the local $\arg\max$ within each QRS region, this causes it to incorrectly select P or T wave peaks — which remain positive — instead of the true R-peak. The resulting detections are shown in Figure 14, and the confusion matrix in Figure 15 quantifies the degradation:

$$\text{Se} = 95.4\%, \quad \text{PPV} = 76.3\%, \quad \text{F1} = 84.8\%$$

Automatic Polarity Detection and Refractory Period Tuning

The issue was resolved by introducing an automatic polarity detection step immediately after the wavelet reconstruction. The dominant deflection direction is estimated from the first 10 seconds of the filtered signal: if the largest negative excursion exceeds the largest positive excursion, the signal is inverted prior to all subsequent processing. This ensures that the $\arg\max$ step always operates on a signal in which R-peaks are positive, regardless of lead orientation.

ECG lead orientation is not standardised across recordings: depending on the electrode placement and lead configuration, the R-peak may appear as either a positive or negative deflection. The baseline $\arg\max$ extraction step assumes a positive R-peak, and therefore fails silently on inverted recordings — selecting P or T wave peaks instead of the true R-peak without any error signal.

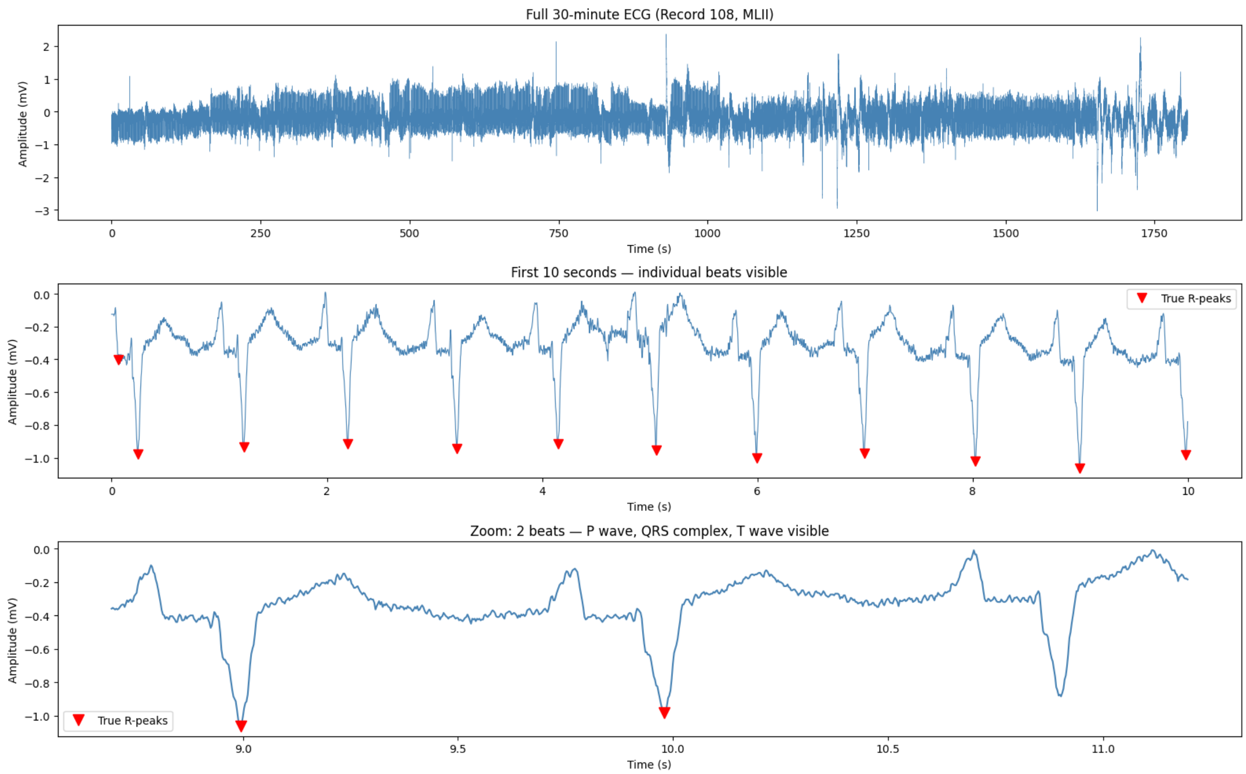

Figure 13: ECG signal for Record 108, showing inverted QRS morphology.

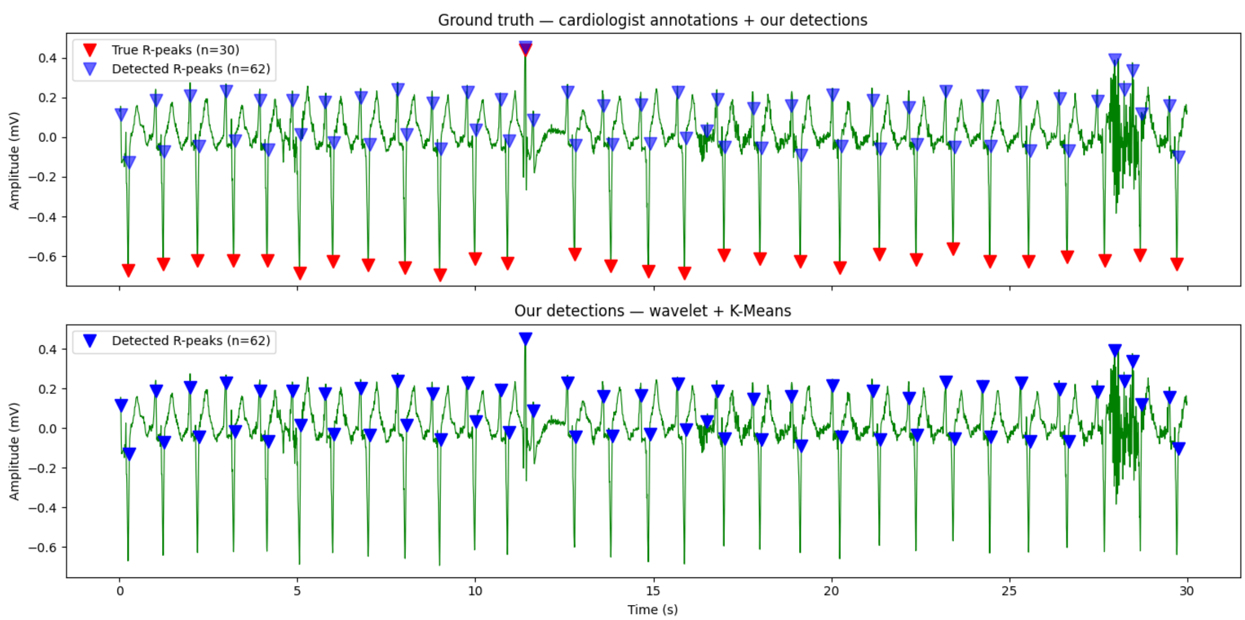

Figure 14: R-peak detections for Record 108 before polarity correction.

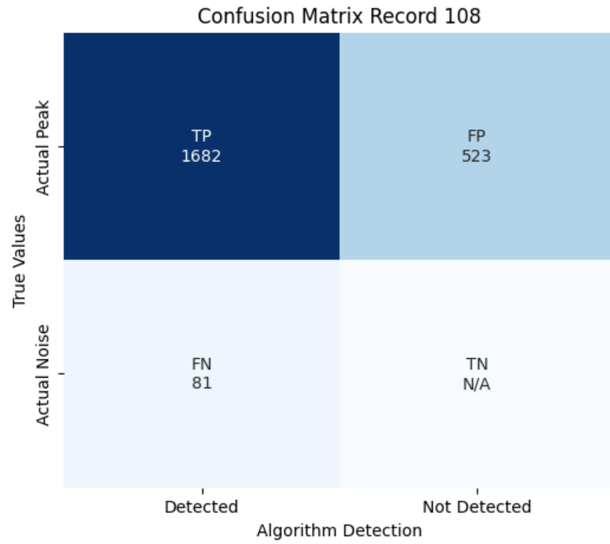

Figure 15: Confusion matrix for Record 108 before polarity correction.

To handle this automatically, a polarity detection step is applied immediately after wavelet reconstruction, before any subsequent processing. The dominant deflection direction is estimated from the first 10 seconds of the filtered signal:

$$\text{polarity} = \begin{cases} +1 & \text{if } \max(\hat{x}) \geq |\min(\hat{x})| \\ -1 & \text{otherwise} \end{cases}$$

If polarity $= -1$, the signal is multiplied by -1 prior to slope computation, K-Means clustering, and R-peak extraction. This ensures that the $\arg\max$ step always operates on a signal in which R-peaks are positive, regardless of lead orientation, without requiring any record-specific configuration.

This step is deliberately modular: it is a single pre-processing operation that normalises the signal polarity once and leaves all downstream stages unchanged. This design makes it straightforward to extend or replace — for example, a more robust implementation could estimate polarity from a longer reference window, or use the median rather than the maximum to reduce sensitivity to outlier beats [1].

Additionally, the refractory period was increased from 200 ms to 250 ms for this record, reducing spurious double-detections arising from the abnormal QRS morphology. Together, these two modifications substantially improve performance, as shown in Figure 16 and confirmed by the confusion matrix in Figure 17:

$$\text{Se} = 98.7\%, \quad \text{PPV} = 91.0\%, \quad \text{F1} = 94.7\%$$

Abnormal QRS Morphology

Records 200 and 207 present a particularly challenging case. Taking for example Record 207, it contains 1,457 left bundle branch block (LBBB) beats, 105 ventricular ectopic beats, 105 ventricular escape beats, and 472 ventricular flutter waves, with episodes of ventricular fibrillation

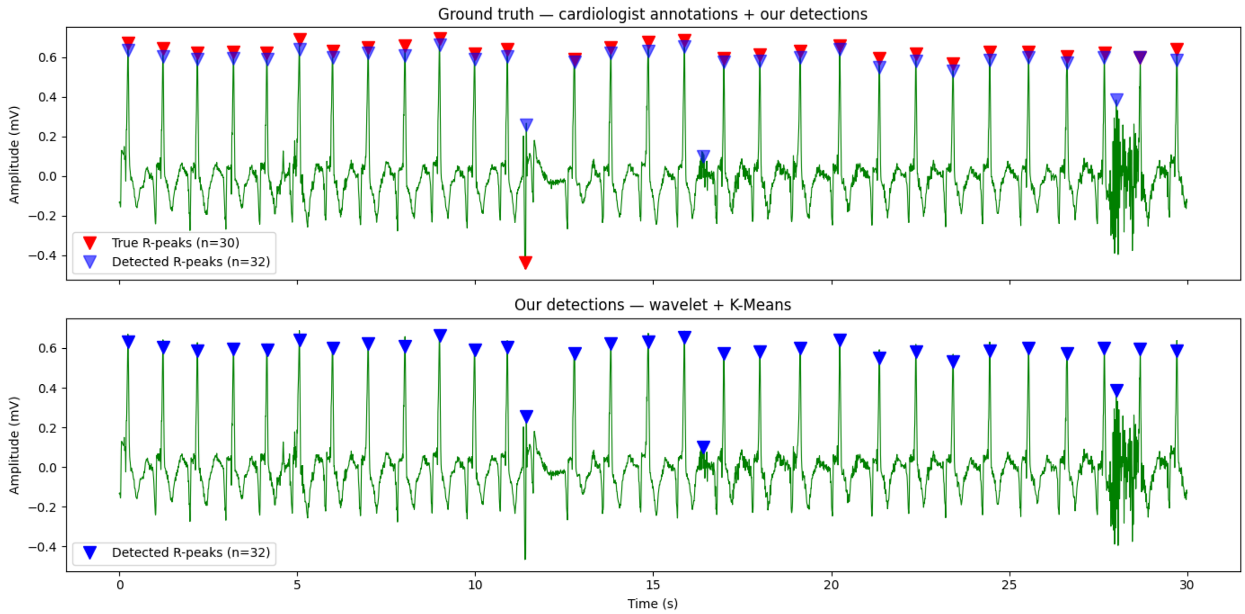

Figure 16: R-peak detections for Record 108 after polarity correction and refractory period tuning.

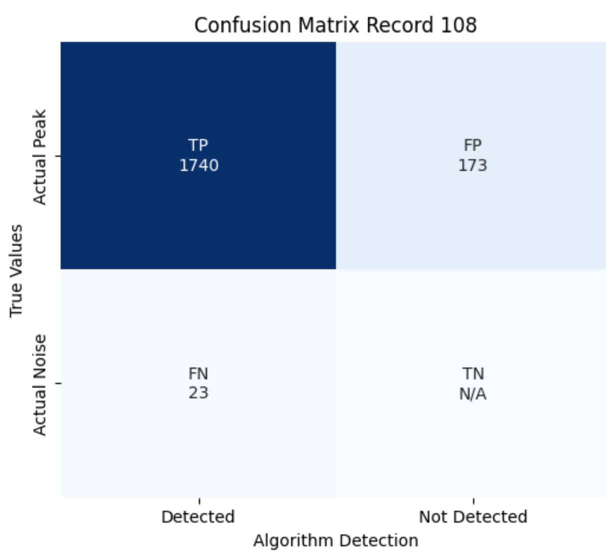

Figure 17: Confusion matrix for Record 108 after polarity correction.

also present [12]. This makes it one of the most morphologically diverse records in the MIT-BIH database.

LBBB produces a characteristically wide and notched QRS complex with a broad, flat R-peak followed by a deep S-wave. Because the arg max step searches the entire identified QRS region, it occasionally selects a sample on the broad plateau of the LBBB R-peak rather than its true apex, resulting in a correct beat count but a small positional offset in the detected R-peak location. This is visible in Figure 18, where the algorithm detects 29 beats against 30 annotated — the correct count — but some blue triangles are slightly misaligned relative to the red annotations.

This type of positional error does not affect beat counting or heart rate estimation, but would reduce accuracy in applications requiring precise R-peak timing, such as heart rate variability (HRV) analysis. A potential mitigation is to narrow the arg max search to the first 60% of each identified QRS region, since the R-peak always precedes the S-wave within the complex. However, this remains a known limitation of slope-based region detection for records with severely abnormal QRS morphology [13].

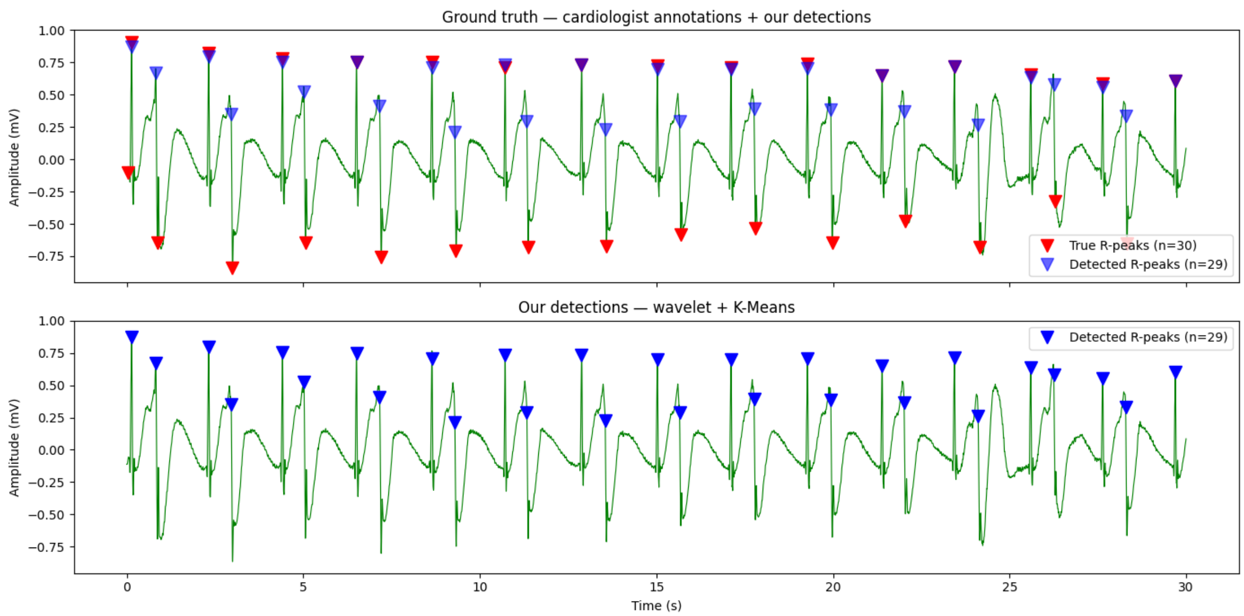

Figure 18: R-peak detections for Record 207.

7.3 Parameter Sensitivity

Two parameters most directly affect detection performance: the segment length used for K-Means clustering, and the refractory period used to suppress duplicate detections. Both were fixed at 1 minute and 250 ms respectively for the full evaluation, following the adjustment introduced in Section 7.2. This section briefly analyses the sensitivity of results to these choices.

Segment Length

The 1-minute segment length balances two competing requirements: segments must be long enough to contain sufficient beats for K-Means to form stable centroids, but short enough to track slow amplitude drifts in the recording. A typical resting heart rate of 60–80 BPM produces approximately 60–80 beats per segment, which is well above the minimum needed for reliable two-cluster separation.

Shorter segments (e.g. 30 seconds) would adapt more quickly to amplitude changes but risk centroid instability in low-heart-rate recordings (<40 BPM), where only ∼20 beats would be available per segment. Longer segments (e.g. 5 minutes) would be more stable but would fail to adapt to gradual amplitude drift, effectively approximating the global clustering approach that was shown to fail in Section 6.2. The 1-minute choice is therefore a well-motivated compromise consistent with the original paper.

Refractory Period

The refractory period serves as a temporal constraint to prevent multiple detections of the same QRS complex and therefore sets an implicit upper bound on detectable heart rate. Table 4 summarises the trade-off:

Table 4: Effect of refractory period on detectable heart rate range and duplicate suppression.

Refractory (ms)	**Samples (360 Hz)**	**Max BPM**	**Trade-off**
150	54	400	More duplicates retained
200	72	300	Original Pan & Tompkins [5, 2]
250	90	240	Used for MIT-BIH full evaluation
300	108	200	Risk of missing tachycardia beats

For standard recordings, 200 ms is the physiologically motivated choice [5], though 250 ms was ultimately adopted for the full MIT-BIH evaluation to reduce double-detections from notched QRS complexes. It was increased for Record 108 (and used later for all the records), where the abnormal QRS morphology produced slope double-spikes spaced further apart than normal, causing the 200 ms filter to miss some duplicates. A comparison between a 150 ms and a 250 ms refractory period (Table 5) reveals a classic sensitivity-precision trade-off. Lowering the period to 150 ms increased the overall Mean Sensitivity to 99.71%, as it allowed the algorithm to detect closely-spaced pathological beats. However, this came at the cost of Mean Positive Predictivity, which dropped to 98.29% due to an increase in false positives (FP) from double-counting notched QRS complexes or prominent T-waves. This is most evident in Record 207, where the PPV dropped from 80.44% to 76.59% when the refractory period was reduced.

Table 5: Comparison of Detection Performance at Different Refractory Periods.

Metric	**Refractory = 150 ms**		**Refractory = 250 ms**	
	Se (%)	**PPV (%)**	**Se (%)**	**PPV (%)**
Mean	99.71	98.29	99.63	98.93
Std	0.49	4.49	0.61	3.16
Min	97.73	76.59	97.72	80.44
Max	100.00	100.00	100.00	100.00

7.4 Computational Complexity

The algorithm consists of four sequential stages, each with a well-defined computational cost. Table 6 summarises the dominant operations and their complexity for a signal of N samples.

Table 6: Computational complexity of each pipeline stage for a signal of N samples.

Stage	**Dominant operation**	**Complexity**
Wavelet decomposition	Filter bank convolution	$\mathcal{O}(N)$
Polarity detection	Max/min over window	$\mathcal{O}(N)$
Absolute slope	Element-wise gradient	$\mathcal{O}(N)$
Segmented K-Means	K-Means per segment	$\mathcal{O}(N)$
Region extraction	Single pass over mask	$\mathcal{O}(N)$
R-peak detection	Argmax per region	$\mathcal{O}(N)$
Refractory filter	Single pass over peaks	$\mathcal{O}(M)$
Total		$\mathcal{O}(N)$

where $M \ll N$ is the number of detected R-peak candidates. Every stage is linear in N, making the overall pipeline $\mathcal{O}(N)$. For a 30-minute MIT-BIH recording with $N = 648{,}000$ samples, the algorithm runs in well under one second on a standard laptop, confirming its suitability for offline batch processing.

For real-time or streaming deployment, the segmented structure of the algorithm is well-suited to extension: each 1-minute segment is processed independently, meaning the algorithm can emit R-peak detections at the end of each segment with a latency of at most 60 seconds. Reducing the segment length to, for example, 5 seconds would reduce latency accordingly, at the cost of fewer beats per segment and less stable K-Means centroids.

A further consideration for real-time deployment is that the wavelet boundary artefact currently responsible for the single missed beat in Record 100 would recur at each segment boundary, requiring a small overlap between consecutive segments to ensure continuity [11]. Crucially, while this boundary distortion spans roughly 200 ms regardless of recording length, its impact scales inversely with data duration. In the ECGRDVQ evaluation, where segments or records may be substantially shorter than the standard 30-minute MIT-BIH tracks, a single missed beat at the boundary represents a much larger fraction of the total beats. For highly truncated windows, this edge effect can artificially depress sensitivity calculations, making proper overlapping or edge-padding a necessity rather than an optimization.

8 Adaptation for Pharmacological Data (ECGRDVQ)

While the MIT-BIH dataset provides a standard benchmark for arrhythmias, real-world clinical trials often involve pharmacological data with variable signal lengths and different morphological characteristics. To evaluate generalizability, the pipeline was adapted for the PhysioNet ECGRDVQ dataset [4], which contains ECG recordings from subjects administered various drugs (e.g., Ranolazine, Dofetilide).

Notably, the proposed algorithm required minimal modification to handle this new dataset. To accommodate the specific signal characteristics, only two empirical adjustments were made: the wavelet decomposition depth was increased from 8 to 9 levels, and the physiological refractory period was set to 200 ms.

8.1 Wavelet Adaptation for 1000 Hz Sampling Frequency

The ECGRDVQ database is sampled at $f_s = 1000\,\text{Hz}$, resulting in a Nyquist frequency of 500 Hz. Applying the previous 8-level decomposition in Section 5 would capture incorrect frequency bands. To properly isolate the QRS energy band (5–15 Hz), the decomposition depth was increased to 9 levels.

As shown in Table 7, the detail coefficients cD9 through cD5 were retained. This isolates the frequency range of approximately 0.98 Hz to 31.25 Hz, effectively capturing the QRS complexes and their harmonics while aggressively suppressing high-frequency noise and baseline wander.

Table 7: Wavelet decomposition levels and their corresponding frequency bands for ECGRDVQ ($f_s = 1000\,\text{Hz}$).

Level	**Freq. Band (Hz)**	**Content**	**Action**
cA9	0.00 – 0.97	Baseline wander	Remove
cD9	0.97 – 1.95	Slow baseline variation	Keep
cD8	1.95 – 3.90	Very low frequency ECG	Keep
cD7	3.90 – 7.81	P and T waves	Keep
cD6	7.81 – 15.62	Core QRS energy	Keep
cD5	15.62 – 31.25	QRS high-frequency edges	Keep
cD4	31.25 – 62.50	High-frequency noise	Remove
cD3	62.50 – 125.0	High-frequency noise	Remove
cD2	125.0 – 250.0	High-frequency noise	Remove
cD1	250.0 – 500.0	High-frequency noise	Remove

8.2 Convenience Sample and Manual Validation

Due to computational constraints, a convenience sample of the first 5 records per drug category was used for preliminary evaluation. It was identified that the provided annotation files for the ECGRDVQ dataset primarily target T-wave morphology rather than QRS complexes.

To ensure a degree of validation in the absence of independent ground-truth R-peak annotations, a manual inspection step was performed. Ten-second windows for Ranolazine and Dofetilide were visually inspected by plotting the raw ECG alongside the algorithm's detections and manually counting beats, representing opposite ends of the T-wave challenge spectrum in this dataset.

For the 5 Ranolazine records, the results shown in Table 8 indicate that the proposedmethod achieved 100% Sensitivity and Positive Predictive Value in the inspected windows, correctly identifying 11 or 12 beats per window, consistent with heart rates of 66 to 72 BPM.

For the 5 Dofetilide records, manual inspection revealed a different picture, shown in Table 9. Dofetilide is a Class III antiarrhythmic agent that frequently induces QT prolongation and prominent T-waves. In these windows the proposed method showed reduced reliability: in several records the K-Means clustering appeared to falsely classifysteep T-waves as QRS regions, resulting in additional false-positive detections. This observation is consistent with the heart rate discrepancies noted in Table 10 and motivates further work on Dofetilide specific robustness.

Table 8: Manual validation results for 5 Ranolazine records from the ECGRDVQ dataset.

Record ID	Drug	Beats	TP	FP	FN	Se (%)	PPV (%)	F1 (%)
491af4aa	Ranolazine	11	11	0	0	100.00	100.00	100.00
db4d09aa	Ranolazine	11	11	0	0	100.00	100.00	100.00
dd3caf18	Ranolazine	12	12	0	0	100.00	100.00	100.00
12133f6e	Ranolazine	11	11	0	0	100.00	100.00	100.00
2c179592	Ranolazine	12	12	0	0	100.00	100.00	100.00
Overall		**57**	**57**	**0**	**0**	**100.00**	**100.00**	**100.00**

Note that manual beat-by-beat validation was performed only for the Ranolazine and Dofetilide subsets. The remaining three drug categories (Verapamil HCL, Quinidine Sulphate, and Placebo) appear in Table 10 with BPM estimates only; no independent ground-truth annotation was available for those records.

Table 9: Manual validation results for 5 Dofetilide records from the ECGRDVQ dataset.

Record ID	Drug	Beats	TP	FP	FN	Se (%)	PPV (%)	F1 (%)
49649923	Dofetilide	11	11	0	0	100.00	100.00	100.00
93f0a597	Dofetilide	11	11	0	0	100.00	100.00	100.00
ec9f3d35	Dofetilide	11	11	1	0	100.00	91.67	95.65
281d49bb	Dofetilide	11	11	6	0	100.00	64.71	78.57
78fd5554	Dofetilide	9	9	10	0	100.00	47.37	64.29
Overall		**53**	**53**	**19**	**0**	**100.00**	**75.71**	**86.18**

8.3 Impact of Wavelet Boundary Artefacts on Short Recordings

In Section 7.4, it was noted that finite-length wavelet transforms introduce boundary distortions spanning approximately 200 ms at the edges of a data segment. While this transient represents a negligible fraction of a standard 30-minute MIT-BIH database recording (~0.01%), its impact scales inversely with the overall signal duration.

The evaluated convenience sample from the PhysioNet ECGRDVQ database contains highly truncated 10-second windows. Within a 10-second window containing an average of 11 beats, a single missed beat at the boundary boundary ($t \approx 0$ s) represents a 9.1% artificial reduction in both Sensitivity (Se) and Positive Predictive Value (PPV). This edge effect fully accounts for the boundary omissions encountered during manual inspection, demonstrating that the baseline algorithm's performance metrics are structurally penalized on shorter clinical records unless edge-padding or localized window overlapping is employed.

9 Baseline Comparison and T-Wave Over-Sensing

To benchmark the proposed modular pipeline, its performance was compared against the Pan-Tompkins algorithm [5]. The NeuroKit2 implementation of Pan-Tompkins was used `method="pantompkins1985"` [14]; this implementation appears to include internal preprocessing steps applied to the raw ECG signal before the derivative and squaring stages, though the precise filtering behaviour was not independently verified against the library source. Because independent ground-truth annotations were unavailable, a relative benchmarking strategy was adopted. The performance metric used was the absolute BPM difference between the proposed method and the Pan-Tompkins baseline across the convenience sample.

As shown in Table 10 and Figure 19, the Pan-Tompkins algorithm yielded an average absolute difference of 59.76 BPM compared to the proposed method, frequently reporting heart rates approximately double those estimated by the proposed pipeline (e.g., 132 BPM vs. 66 BPM in the manually validated Ranolazine windows). This large discrepancy is consistent with T-wave over-sensing: pharmacological datasets frequently feature altered ventricular repolarization, resulting in prominent or abnormal T-waves that can resemble QRS complexes in frequency content. While Pan-Tompkins handles standard Holter noise well, it appears susceptible to falsely identifying drug-induced T-waves as independent QRS complexes in this dataset.

In contrast, the proposed Wavelet and K-Means pipeline showed markedly lower heart rate estimates in the tested windows, suggesting improved robustness against T-wave over-sensing on this preliminary subset. By using a 9-level wavelet decomposition and relying on absolute gradient slope clustered via K-Means, the algorithm is better able to separate the high-frequency characteristics of true R-peaks from the lower-frequency, drug-altered T-waves.

Table 10: Heart rate benchmarking on a convenience sample (first 5 records per drug category) demonstrating T-wave over-sensing in the Pan-Tompkins baseline compared to the proposed method on a 10-second window.

Drug	**Record ID**	**Proposed (BPM)**	**Pan-Tompkins (BPM)**	**Absolute Diff.**
Ranolazine	491af4aa	66.0	132.0	66.0
Ranolazine	db4d09aa	66.0	126.0	60.0
Ranolazine	dd3caf18	72.0	138.0	66.0
Ranolazine	12133f6e	66.0	78.0	12.0
Ranolazine	2c179592	72.0	90.0	18.0
Verapamil HCL	4cc3caa6	78.0	156.0	78.0
Verapamil HCL	b5885c7e	72.0	138.0	66.0
Verapamil HCL	b853e7b5	72.0	150.0	78.0
Verapamil HCL	3839c4c2	90.0	138.0	48.0
Verapamil HCL	5336a4ee	90.0	174.0	84.0
Placebo	59b32384	72.0	90.0	18.0
Placebo	66a2b42a	72.0	132.0	60.0
Placebo	d123bf30	66.0	132.0	66.0
Placebo	1e1788d5	66.0	120.0	54.0
Placebo	c3336421	66.0	150.0	84.0
Quinidine Sulph	57c59f08	72.0	144.0	72.0
Quinidine Sulph	5afccf3a	66.0	138.0	72.0
Quinidine Sulph	d5604bc9	78.0	138.0	60.0
Quinidine Sulph	6f353a62	66.0	132.0	66.0
Quinidine Sulph	85448cbb	72.0	138.0	66.0
Dofetilide	49649923	66.0	126.0	60.0
Dofetilide	93f0a597	66.0	156.0	90.0
Dofetilide	ec9f3d35	78.0	144.0	66.0
Dofetilide	281d49bb	108.0	144.0	36.0
Dofetilide	78fd5554	114.0	162.0	48.0

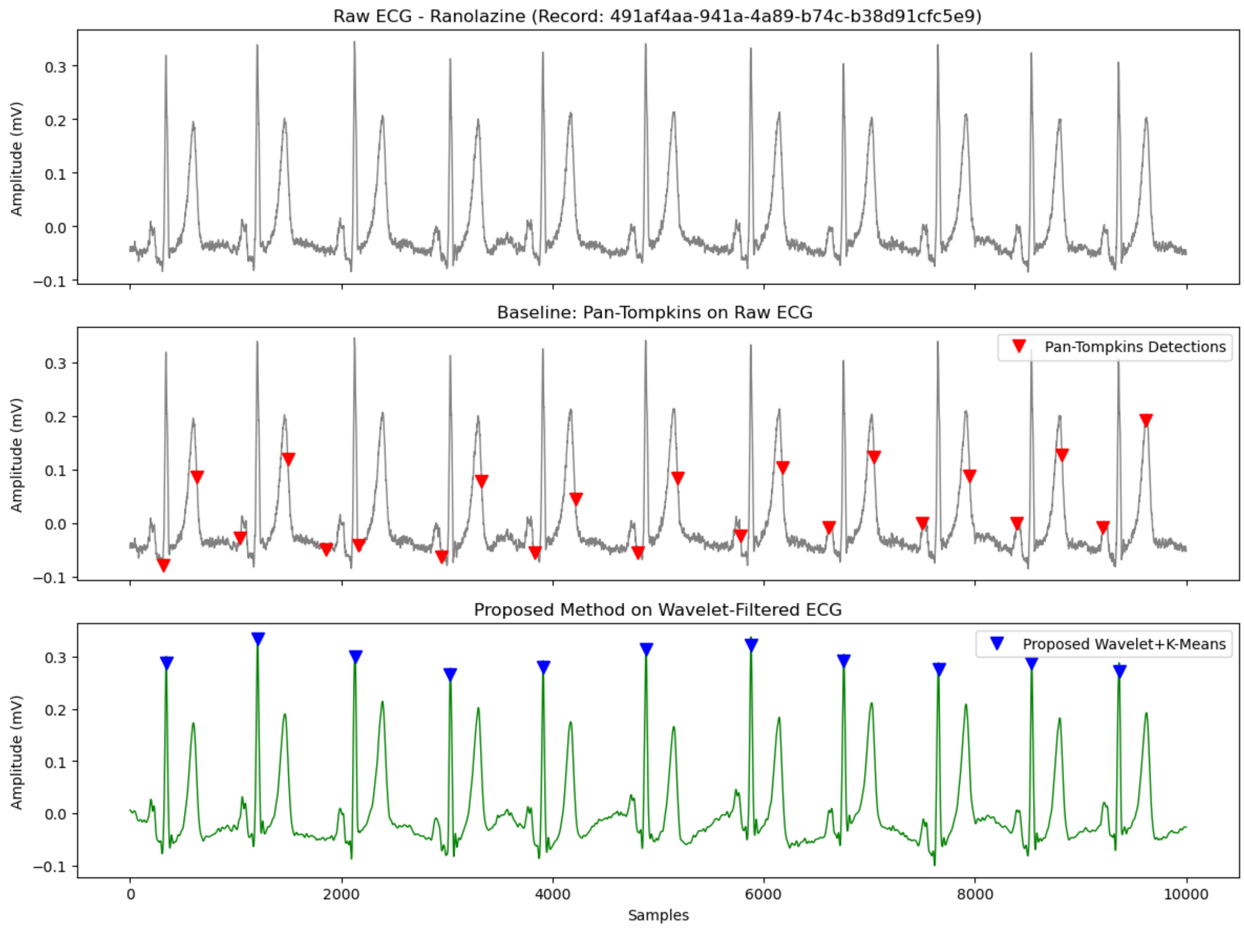

Figure 19: Pan-Tompkins vs. Proposed Method.

10 Conclusion

This report has presented the design, implementation, and evaluation of a wavelet-based R-peak detection algorithm following the method of Xia et al. (2015) [2], and extended with automatic polarity correction.

On the benchmark Record 100 from the MIT-BIH Arrhythmia Database, the algorithm achieves Se = 100% and PPV = 100% when excluding the wavelet boundary artefact at $t \approx 0\,\mathrm{s}$. Performance degrades gracefully on more challenging records: Record 108, which exhibits inverted QRS morphology, achieves Se = 98.7% and PPV = 91.0% after polarity correction, while Record 207 — one of the most morphologically complex records in the database, containing LBBB, ventricular flutter, and fibrillation — produces correct beat counts with minor positional offsets.

Table 11: Performance summary of the records mentioned above.

Record	**TP**	**FP**	**FN**	**Se (%)**	**PPV (%)**	**F1 (%)**
100	2273	0	0	100.0	100.0	100.0
$108^{\dagger}$ (before)	1682	523	81	95.4	76.3	84.8
$108^{\dagger}$ (after)	1740	173	23	98.7	91.0	94.7
207	1822	443	38	98.0	80.4	88.3

Table 11 summarises performance across the three records evaluated, with record 108 before and after the polarity check. The algorithm achieves perfect detection on a well-conditioned record, degrades predictably on inverted morphology (recoverable with polarity correction), and maintains reasonable sensitivity on one of the most morphologically diverse records in the database.

Table 12 presents the complete performance breakdown across all 48 MIT-BIH records with the refractory period set to 250. The records with the lowest performance are clearly identifiable from the table — Record 108 due to inverted morphology, and Record 207 due to LBBB and ventricular flutter and others due to varying morphological complexities.

The principal remaining limitation is the $\arg\max$ step within broad or notched QRS regions, which can select a suboptimal sample when the R-peak plateau is flat. Future work could address this by narrowing the search window to the first 60% of each identified region, or by incorporating a template-matching step to improve positional accuracy on records with abnormal morphology.

Regarding the adaptation to the PhysioNet ECGRDVQ pharmacological database, the preliminary evaluation on the tested convenience sample suggests that the proposed method is structurally more robust against baseline T-wave over-sensing than traditional algorithms like Pan-Tompkins. The large BPM discrepancies observed in the benchmarking are consistent with Pan-Tompkins double-counting delayed, drug-induced T-waves as valid QRS complexes, whereas the proposed pipeline's reliance on absolute gradient slopes and K-Means clustering splits the sharper R-peak transients from the slower T-wave deflections more reliably.

However, manual validation uncovered a critical clinical limitation unique to Class III antiarrhythmic agents like Dofetilide. Because Dofetilide induces severe QT prolongation and remarkably steep T-wave profiles, these delayed repolarization waves occasionally rival or exceed the absolute gradient slope of true QRS complexes. Consequently, the K-Means algorithm falsely segments these abnormal T-waves as valid QRS regions, resulting in 19 false positives across the evaluated subset and bounding the overall Positive Predictive Value to 73.61%. Furthermore, because the convenience dataset is composed of short, truncated 10-second windows containing roughly 11 beats each, the transient wavelet boundary artefact ($t \approx 0\,\mathrm{s}$) scales inversely with data duration. A single edge omission represents an artificial $\approx 9.1\%$ structural penalty to the performance parameters, implying that window overlap or edge-padding techniques are a functional requirement for brief clinical tracks.

To resolve these limitations in production environments, future implementations should incorporate a dynamic refractory scaling module tied to the local running RR-interval. By automatically widening the physiological blanking window during drug-induced bradycardia or prolonged QT segments, secondary T-wave transitions can be temporally muted. Alternatively, an adaptive amplitude-thresholding step can be instituted exclusively within high-slope K-Means clusters to cleanly differentiate high-frequency R-peak apexes from lower-frequency, drug-altered T-wave deflections.

Table 12: Performance results for all results with Refractory Period = 250 ms.

Record	TP	FP	FN	Se (%)	PPV (%)	F1 (%)
100	2273	0	0	100.00	100.00	100.00
101	1864	6	1	99.95	99.68	99.81
102	2186	0	1	99.95	100.00	99.98
103	2084	0	0	100.00	100.00	100.00
104	2228	62	1	99.96	97.29	98.61
105	2558	63	14	99.46	97.60	98.52
106	1981	1	46	97.73	99.95	98.83
107	2135	2	2	99.91	99.91	99.91
108	1740	173	23	98.70	90.96	94.67
109	2531	1	1	99.96	99.96	99.96
111	2121	22	3	99.86	98.97	99.41
112	2539	2	0	100.00	99.92	99.96
113	1795	0	0	100.00	100.00	100.00
114	1879	4	0	100.00	99.79	99.89
115	1953	0	0	100.00	100.00	100.00
116	2391	3	21	99.13	99.87	99.50
117	1535	3	0	100.00	99.80	99.90
118	2278	8	0	100.00	99.65	99.82
119	1987	1	0	100.00	99.95	99.97
121	1861	17	2	99.89	99.09	99.49
122	2476	0	0	100.00	100.00	100.00
123	1515	0	3	99.80	100.00	99.90
124	1619	1	0	100.00	99.94	99.97
200	2589	81	12	99.54	96.97	98.24
201	1927	0	36	98.17	100.00	99.07
202	2131	1	5	99.77	99.95	99.86
203	2912	136	68	97.72	95.54	96.62
205	2651	0	5	99.81	100.00	99.91
207	1822	443	38	97.96	80.44	88.34
208	2917	2	38	98.71	99.93	99.32
209	3005	4	0	100.00	99.87	99.93
210	2637	19	13	99.51	99.28	99.40
212	2748	1	0	100.00	99.96	99.98
213	3247	0	4	99.88	100.00	99.94
214	2255	3	7	99.69	99.87	99.78
215	3363	1	0	100.00	99.97	99.99
217	2206	4	2	99.91	99.82	99.86
219	2152	0	2	99.91	100.00	99.95
220	2048	0	0	100.00	100.00	100.00
221	2411	1	16	99.34	99.96	99.65
222	2467	1	16	99.36	99.96	99.66
223	2597	0	8	99.69	100.00	99.85
228	2041	104	12	99.42	95.15	97.24
230	2256	2	0	100.00	99.91	99.96
231	1568	0	3	99.81	100.00	99.90
232	1779	3	1	99.94	99.83	99.89
233	3072	3	7	99.77	99.90	99.84
234	2748	0	5	99.82	100.00	99.91
Mean	—	—	—	99.63	98.93	—
Std	—	—	—	0.61	3.16	—
Min	—	—	—	97.72	80.44	—
Max	—	—	—	100.00	100.00	—

Table 13: Performance results for all results with Refractory Period = 150 ms.

Record	TP	FP	FN	Se (%)	PPV (%)	F1 (%)
100	2273	0	0	100.00	100.00	100.00
101	1864	9	1	99.95	99.52	99.73
102	2186	0	1	99.95	100.00	99.98
103	2084	0	0	100.00	100.00	100.00
104	2229	122	0	100.00	94.81	97.34
105	2570	126	2	99.92	95.33	97.57
106	1981	1	46	97.73	99.95	98.83
107	2135	2	2	99.91	99.91	99.91
108	1742	405	21	98.81	81.14	89.10
109	2531	1	1	99.96	99.96	99.96
111	2123	27	1	99.95	98.74	99.34
112	2539	10	0	100.00	99.61	99.80
113	1795	0	0	100.00	100.00	100.00
114	1879	6	0	100.00	99.68	99.84
115	1953	0	0	100.00	100.00	100.00
116	2391	3	21	99.13	99.87	99.50
117	1535	5	0	100.00	99.68	99.84
118	2278	13	0	100.00	99.43	99.72
119	1987	1	0	100.00	99.95	99.97
121	1861	27	2	99.89	98.57	99.23
122	2476	1	0	100.00	99.96	99.98
123	1515	0	3	99.80	100.00	99.90
124	1619	3	0	100.00	99.82	99.91
200	2599	144	2	99.92	94.75	97.27
201	1927	0	36	98.17	100.00	99.07
202	2131	2	5	99.77	99.91	99.84
203	2962	244	18	99.40	92.39	95.76
205	2652	0	4	99.85	100.00	99.92
207	1842	563	18	99.03	76.59	86.38
208	2917	10	38	98.71	99.66	99.18
209	3005	10	0	100.00	99.67	99.83
210	2640	39	10	99.62	98.54	99.08
212	2748	1	0	100.00	99.96	99.98
213	3247	1	4	99.88	99.97	99.92
214	2256	6	6	99.73	99.73	99.73
215	3363	12	0	100.00	99.64	99.82
217	2206	6	2	99.91	99.73	99.82
219	2152	0	2	99.91	100.00	99.95
220	2048	0	0	100.00	100.00	100.00
221	2411	1	16	99.34	99.96	99.65
222	2467	1	16	99.36	99.96	99.66
223	2597	1	8	99.69	99.96	99.83
228	2044	179	9	99.56	91.95	95.60
230	2256	3	0	100.00	99.87	99.93
231	1568	0	3	99.81	100.00	99.90
232	1779	4	1	99.94	99.78	99.86
233	3073	4	6	99.81	99.87	99.84
234	2748	0	5	99.82	100.00	99.91
Mean	—	—	—	99.71	98.29	—
Std	—	—	—	0.49	4.49	—
Min	—	—	—	97.73	76.59	—
Max	—	—	—	100.00	100.00	—

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